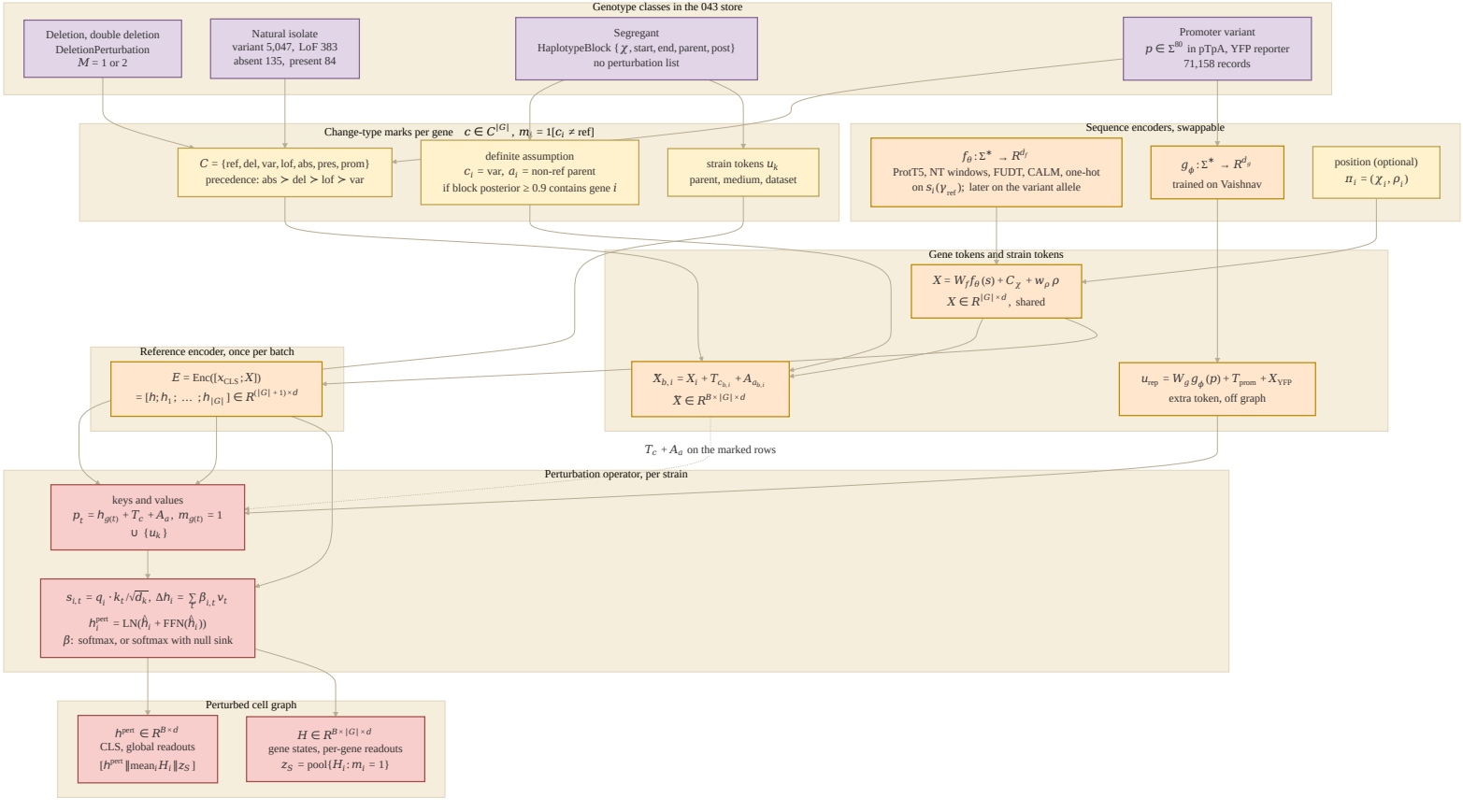

Genotype classes in the 043 store
Deletion, double deletion
DeletionPerturbation
M = 1 or 2
Natural isolate
variant 5,047, LoF 383
absent 135, present 84
Segregant
HaplotypeBlock { χ, start, end, parent, post}
no perturbation list
Promoter variant
$p \in \Sigma^{80}$ in pTpA, YFP reporter
71,158 records
Change-type marks per gene $c \in C^{|G|}$, $m_i = 1[c_i \neq \text{ref}]$
C = {ref, del, var, lof, abs, pres, prom}
precedence: abs ≻ del ≻ lof ≻ var
definite assumption
c_i = var, a_i = non-ref parent
if block posterior ≥ 0.9 contains gene i
strain tokens u_k
parent, medium, dataset
Sequence encoders, swappable
$f_\theta : \Sigma^* \to R^{d_f}$
ProtT5, NT windows, FUDT, CALM, one-hot
on $s_i(\gamma_{\text{ref}})$; later on the variant allele
$g_\phi : \Sigma^* \to R^{d_g}$
trained on Vaishnav
position (optional)
$\pi_i = (\chi_i, \rho_i)$
Gene tokens and strain tokens
$X = W_f f_\theta(s) + C_\chi + w_\rho \rho$
$X \in R^{|G| \times d}$, shared
Reference encoder, once per batch
$E = \text{Enc}([x_{\text{CLS}}; X])$
$= [h; h_1; \ldots; h_{|G|}] \in R^{(|G|+1) \times d}$
$X_{b,i} = X_i + T_{c_{b,i}} + A_{a_{b,i}}$
$X \in R^{B \times |G| \times d}$
$u_{\text{rep}} = W_g g_\phi(p) + T_{\text{prom}} + X_{\text{YFP}}$
extra token, off graph
$T_c + A_a$ on the marked rows
Perturbation operator, per strain
keys and values
$p_t = h_{g(t)} + T_c + A_a$, $m_{g(t)} = 1$
$\cup \{u_k\}$
$s_{i,t} = q_i \cdot k_t / \sqrt{d_k}$, $\Delta h_i = \sum_t \beta_{i,t} v_t$
$h_i^{\text{pert}} = \text{LN}(\hat{h}_i + \text{FFN}(\hat{h}_i))$
β: softmax, or softmax with null sink
Perturbed cell graph
$h^{\text{pert}} \in R^{B \times d}$
CLS, global readouts
$[h^{\text{pert}} \| \text{mean}_i H_i \| z_S]$
$H \in R^{B \times |G| \times d}$
gene states, per-gene readouts
$z_S = \text{pool}\{H_i : m_i = 1\}$